## The Threat to India

**Globally, the financial sector has been among the fastest adopters of AI tools.** In India, the use-cases range from conversational AI for payments, email and workflow automation, as well as business intelligence. GenAI is set to improve Indian banking operations by up to 46 per cent, according to a 2025 EY report. However, the greater the AI integration, the greater the risks to market integrity, consumer trust and systemic vulnerabilities, all of which need to be well understood. In a hypothetical worst-case scenario, an AI-powered attack could overwhelm banks and payment networks with convincing fraudulent transactions through fake identities and authorisations. AI could also break into several interconnected financial systems, corrupting transaction records and leaving banks in disarray.

Its other major threat could be to national security, external and internal. AI could poison the information feeding military command systems, alter satellite images and contaminate intelligence reports. Air-defence systems could misidentify targets, drones attack the wrong forces, missiles trigger autonomously and cause wars in which commanders can't tell between authentic orders and fake ones. During an election, terror attack or communal riot, rogue agents could plant thousands of convincing videos, voice messages and social media accounts.

AI chaos could well extend to infrastructure. India's infrastructure systems are acquiring an AI-powered digital nervous system, promising tremendous gains in efficiency. But the more deeply AI is embedded in critical systems, the greater the potential attack surface. If AI compromises the digital nervous system of transportation, railway signalling could develop unexplained faults, airport systems receive fake flight information and traffic management systems could orchestrate a gridlock. AI could infiltrate the power grid too, unleashing mayhem across cities.

A similar scenario could unfold in industry. AI controlling a highly automated factory could get poisoned. Thus, quality-control cameras could approve defective components, supply-chain systems send the wrong materials to the wrong plants, robots malfunction, chemical processes move outside safe limits, critical components stop reaching customers.

# THE AI LEXICON

### 10 ESSENTIAL TERMS YOU NEED TO KNOW

**1. AI SWARM:** Group of autonomous AI agents that collaborate, share information and divide tasks independently to achieve a common goal

**2. RECURSIVE SELF-IMPROVEMENT:** The process by which AI systems use their own capabilities to design, optimise or develop more capable successor versions

**3. SANDBOX:** An isolated computing environment used to safely test software or AI systems without affecting the wider system

**4. AI ALIGNMENT:** The process of designing AI systems to behave in ways that are consistent with human values, goals and intentions

**5. KILL SWITCH**
A mechanism that can shut down an AI system in an emergency or if it behaves in an unsafe manner



**6. JAILBREAKING**
Techniques, often involving carefully crafted prompts, used to bypass an AI system's safeguards and make it produce restricted content

**7. DATA POISONING**
A cyberattack that corrupts the data used to train an AI model, with the aim of altering its behaviour or degrading its performance

## Defence Vulnerabilities

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